



A C U M E N U S

Z E P H Y R U S P L A T F O R M

OCEL Models for Process Improvement in Modern Hospital Operations

*Research Landscape, Reference Ontology, Model Catalog, and Zephyrus Implementation
Priorities*

ACUM-OPS-OCEL-001

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One governed event fabric, many bounded improvement models



Figure 1. A single governed OCEL event fabric supports bounded domain and improvement models without forcing one universal case notion.

Executive conclusion

The correct target is **not one giant hospital process map**. It is one governed, reusable **object-centric event fabric** from which bounded models are generated for specific improvement questions.

The strongest hospital OCEL portfolio has four layers:

1. **A canonical foundation** for identity, encounter, location, work, resources, and governance.
2. **Reusable interaction patterns** such as request-to-fulfillment, queue-to-assignment, readiness synchronization, movement and occupancy, handoff, split/merge, exception/rework, and intervention-to-outcome.
3. **Domain model slices** for ED, beds, discharge, perioperative services, diagnostics, transport, EVS, staffing, pharmacy, transfer centers, quality, and administrative operations.
4. **Analytical models** for discovery, performance, conformance, comparison, prediction, simulation, organizational analysis, and governed intervention.

The first production portfolio should focus on five object intersections where hospital flow actually fails:

- admitted encounter + placement request + bed + unit;
- discharge-ready encounter + barrier + responsible team + downstream service;
- vacated bed + EVS task + staff/assignment + next placement;
- OR case + OR suite + PACU bay/bed + downstream inpatient capacity;
- transport job + encounter + origin + destination + transporter/vehicle.

These interactions are high-value because a single-case patient log cannot faithfully show which object was ready first, which resource arrived late, whether work was pooled, or whether the apparent delay was merely documentation lag.

The evidence must be interpreted conservatively. Direct healthcare object-centric process-mining evidence is still early: a 2024 study transformed OMOP data and MIMIC-IV into OCELs, and a 2026 single-center study applied OCPM to laparoscopic cholecystectomy. Conventional healthcare process mining has a much larger evidence base across ED care, patient journeys, perioperative workflows, laboratory operations, transport, oncology, and conformance. Therefore, most models in this report are **well-motivated adaptations**, not already validated object-centric hospital products.

For Zephyrus specifically, the current OCEL projection is a useful foundation but is not yet semantically sufficient for trustworthy hospital-wide process improvement. Before expansion, it should correct five material issues:

1. model changing relationships such as encounter-to-bed through temporal association objects or relationship events, not timeless O2O edges;
2. separate bed assignment, physical occupancy, physical departure, vacancy, dirty, cleaning, ready, and blocked states;
3. stop labeling dispatch/assignment timestamps as actual transport pickup;
4. make object-change projection genuinely idempotent with a natural uniqueness constraint;
5. distinguish the current heuristic handoff calculations from formal OPerA replay and the current batch rule checks from object-centric Petri-net or streaming conformance.

1. What “OCEL model” means

The phrase is ambiguous. This report uses four distinct meanings.

Layer	Meaning	Example
OCEL data model	The event, object, attribute, and relationship schema	Encounter, Bed, Placement Request, assign-bed event
Domain model	A bounded hospital operational problem represented in OCEL	ED admission-to-bed placement
Process model	A discovered or authored behavioral representation	OC-DFG, object-centric Petri net, object-centric BPMN

Layer	Meaning	Example
Improvement model	The analysis and decision loop applied to the domain	bottleneck analysis, conformance, prediction, simulation, PDSA

OCEL is an event-data standard, not itself an improvement method. OCEL 2.0 represents events, objects, event/object types, qualified event-to-object relationships, qualified object-to-object relationships, and time-varying object attributes. OCPM techniques consume that data to discover, compare, check, explain, predict, or simulate processes. [OCEL 2.0 standard](#) and [formal specification](#).

1.1 Why hospitals are intrinsically object-centric

A hospital does not have one natural case identifier. One operational event can involve:

- a patient and one or more encounters;
- a request and one or more tasks;
- a bed, room, unit, and facility;
- a procedure case and shared procedural resources;
- one or more staff assignments;
- a specimen, analyzer, and ordered test;
- a transport job, origin, destination, and vehicle;
- an alert, recommendation, action, and improvement intervention.

Flattening this into a patient-only trace can duplicate shared events, create false loops, omit resource-only events, and erase synchronization between lifecycles. Object-centric research describes these as convergence, divergence, and deficiency problems. [Object-Centric Process Mining: Unraveling the Fabric of Real Processes](#).

1.2 Evidence grading used in the catalog

Code	Evidence interpretation
D	Direct peer-reviewed healthcare OCPM/OCEL evidence
H	Healthcare process-mining evidence exists, but is mainly case-centric
S	Supported by healthcare standards, workflow semantics, or established operational practice
A	Architecture-derived opportunity; plausible but requires local validation

Most rows have more than one code. An **H+A** model has real healthcare process-mining precedent but its proposed object-centric design is an adaptation. Evidence codes do not score clinical effectiveness.

1.3 Priority grading

Priority	Meaning
P0	Foundation required before trustworthy model deployment
P1	Highest-value first production portfolio
P2	Strong expansion after P1 data and governance are reliable
P3	Later, specialized, or evidence-limited opportunity

2. Research findings that should govern design

2.1 Use one event fabric, many bounded views

OCEL's benefit is reuse: changing the analytical viewpoint should not require a new extraction each time. That does not mean every object belongs on every canvas. A practical hospital implementation should maintain one governed canonical store and publish bounded views with two to five object types for each question.

2.2 Model objects by lifecycle, not by noun proliferation

Create an object type when the entity has:

- a stable identity;
- a meaningful lifecycle of its own;
- relationships to several events or objects;
- independent performance or conformance questions; or
- a need to split, merge, queue, assign, transfer, or be reused.

Keep a value as an attribute when it is merely descriptive. Acuity, priority, status, service line, payer, and isolation requirement are generally attributes unless a governed business process acts on the concept as an independently identified entity.

2.3 Use association objects for relationships that change over time

The OCEL 2.0 O2O relation contains source, target, and qualifier, but no validity interval. This is adequate for relatively stable facts such as `Bed in Unit` or `Encounter of Patient`. It is unsafe for changing relationships such as:

- encounter occupies bed;
- clinician assigned to unit;
- OR case allocated to suite;
- device attached to patient;
- barrier owned by team;
- transport job assigned to transporter.

Use an association object such as `Bed Stay`, `Staff Assignment`, `Case Room Allocation`, `Device Use`, `Barrier Ownership`, or `Transport Assignment`, and attach begin/end/reassign events to it. This is an inference from the OCEL 2.0 metamodel and relational specification, whose O2O relation is not timestamped. [OCEL 2.0 specification, O2O table](#).

2.4 Preserve hospital time semantics

A single timestamp is rarely enough. Each event should retain, where available:

- `occurred_at`: when the physical or clinical activity happened;
- `effective_at`: when a state became effective;
- `documented_at`: when a human recorded it;
- `source_committed_at`: when the source system committed it;
- `ingested_at`: when the event reached the OCEL pipeline;
- `corrected_at`: when a prior fact was amended;
- `time_basis`: actual, device, inferred, scheduled, or documentation proxy;

- `precision`: exact, minute, interval, date-only, or unknown;
- `source_timezone` and normalized UTC timestamp.

Healthcare process-mining literature repeatedly warns that missing events, incorrect timestamps, and imprecise resource information can make analyses misleading. Documentation time can differ materially from execution time. [Process mining for healthcare: characteristics and challenges](#) and [Using Process Mining in Healthcare](#).

2.5 Treat requests, work, and outcomes as separate objects

FHIR explicitly distinguishes definitions, requests, and events, and notes that requests and resulting events are not one-to-one. A `ServiceRequest` can produce a `Procedure`, `DiagnosticReport`, `Observation`, or other result; `Task` can track administrative fulfillment. This maps naturally into an object-centric pattern:

```
Request → Task/Job → Execution Event → Result/Outcome.
```

Do not collapse these into a single status field. [HL7 FHIR R5 workflow patterns](#) and [ServiceRequest](#).

2.6 Discovery is evidence of behavior, not causation

Process discovery can reveal variants, recurrence, delays, and associations. Conformance can show deviation from a reference model. Performance analysis can localize time at object intersections. None of these alone proves that an observed path caused an outcome or that changing it will improve outcomes.

Improvement claims require prospective designs, balancing measures, intervention fidelity, time-series or other appropriate evaluation, and clinical/operational interpretation. CMS requires an ongoing, hospital-wide, data-driven QAPI program; OCEL can support that program but does not replace its governance. [CMS hospital QAPI condition](#).

2.7 Direct healthcare OCPM evidence remains limited

The 2024 OMOP-to-OCEL study demonstrated a transformation method on MIMIC-IV and generated multiple object-centric healthcare process models. It provides technical feasibility, not proof of operational improvement at deployed hospitals. [Park, Lee, and Cho, 2024](#).

The 2026 laparoscopic cholecystectomy study analyzed 1,186 cases with PM4Py and an OC-DFG/variant approach, reported 92% reference-path adherence, and identified preoperative and PACU/discharge bottlenecks. It is a single-center, procedure-specific retrospective study; the authors also caution that an unusually high ICU classification may reflect local data/case-mix interpretation. [Celik, 2026](#).

The broader case-centric healthcare process-mining literature is much larger. A 2025 systematic review retained 53 primary studies from 2014–2024 and found process discovery to be the most common topic; a 2025 tertiary review synthesized 18 secondary reviews and likewise found discovery/control-flow dominant, with noisy/missing data and visualization clarity remaining major issues. [Aversano et al., 2025](#) and [Santos et al., 2025](#).

3. Canonical hospital object ontology

This is a superset catalog. A bounded model should select only what its question requires.

3.1 Enterprise and place

Object type	Lifecycle or role	Typical stable relationships
Health System / Organization	legal or operational enterprise	contains Facility; owns Service Line
Market / Region	geography and network coordination	contains Facility; served by Transfer Center
Facility	hospital, campus, ASC, clinic	contains Department, Unit, Room
Service Line / Program	governed clinical capability	delivered at Facility; uses Department
Department	organizational responsibility	operates Unit/Room; owns Queue
Unit	operational care area	in Facility; contains Room/Bed
Room	physical space and capability	in Unit; contains Bed/Bay
Bed / Bay / Chair	assignable capacity resource	in Room/Unit
Functional Location	imaging, lab, pharmacy, dock, lounge	in Facility

Facility, service line, department, and physical space should remain distinct. A service line crosses locations; a department owns work; a unit is an operational area; a bed is an assignable resource.

3.2 Patient and care context

Object type	Lifecycle or role	Notes
Patient	longitudinal person identity	tokenize; use only when cross-encounter analysis is required
Episode of Care	multi-encounter program/condition episode	useful for oncology, transplant, maternity, readmission
Encounter	visit-level care context	default patient-flow lens
Admission / Stay	inpatient or observation stay	may be separate from ED encounter depending on source semantics
Procedure / OR Case	scheduled/performed procedural lifecycle	distinct from Encounter
Care Plan / Pathway Instance	governed plan instantiated for a patient	useful for conformance

3.3 Requests, work, and constraints

Object type	Lifecycle or role	Examples
Referral	external/internal request for access	oncology referral, specialty consult
Appointment	scheduled service slot	clinic, imaging, procedure
Order / Service Request	authorized clinical service	lab, imaging, therapy, consult
Task / Job	work item with assignment and execution	transport, EVS, case management
Queue	set of work awaiting service	transfer, imaging, pharmacy
Placement Request	request for capability-matched destination	ED admit, ICU stepdown, transfer
Bed Stay / Placement	temporal association of encounter and bed	assign, occupy, leave, close

Object type	Lifecycle or role	Examples
Barrier	constraint with owner, due time, evidence, resolution	authorization, test, consult, transport
Handoff	transfer of responsibility or information	ED-to-unit, OR-to-PACU

3.4 Resources

Object type	Lifecycle or role	Privacy note
Staff Member	person-level workforce identity	often unnecessary and sensitive
Role / Credential	capability definition	preferably static reference data
Shift	scheduled coverage interval	joins demand/capacity analyses
Staff Assignment	temporal role/unit/work association	safer default object than person identity
Care Team	collective responsibility	membership may change
Device / Asset	pump, wheelchair, scanner, bed, robot	instance-level only if individually tracked
Vehicle	internal/external transport resource	ambulance, wheelchair van
Supply Item / Lot	material or inventory unit	lot-level when traceability matters
Instrument Tray / Set	reusable sterile resource	crucial for perioperative logistics

3.5 Diagnostic, therapeutic, and medication objects

Object type	Lifecycle or role
Specimen	collected, transported, accessioned, processed, stored/disposed
Test / Panel	requested analytical work
Analyzer / Instrument	shared diagnostic resource
Imaging Study	ordered, scheduled, acquired, read, finalized
Consult	requested, accepted, seen, recommendation completed
Therapy Episode	PT/OT/SLP/respiratory treatment request and delivery
Medication Order	prescribed therapy request
Pharmacy Work Item	verification, preparation, dispense, delivery
Medication Administration	administered dose event or object if lifecycle requires
Blood Product Unit	allocated, issued, transported, transfused, returned/wasted

3.6 Governance and improvement

Object type	Lifecycle or role
Alert / Signal	opened, acknowledged, suppressed, escalated, cleared
Recommendation	proposed action with rationale and provenance
Operational Action	assigned, accepted, started, completed, overridden, failed
Huddle / Plan	decision cycle and owned action set

Object type	Lifecycle or role
Incident / Safety Event	report, triage, investigate, mitigate, close
PDSA / Intervention	plan, baseline, intervention, study, adopt/adapt/abandon
Reference Model / Policy Version	normative process used for conformance
Metric Definition	governed numerator, denominator, source, threshold, version
Data Quality Finding	detected, assigned, corrected, reconciled, closed

4. Reusable object-centric interaction patterns

Every domain model below is a composition of these patterns.

Pattern	Required objects	Typical events	Improvement questions
Request-to-fulfillment	Request, Task, Encounter, Performer/Resource	order, accept, schedule, start, complete, result, cancel	Where does authorized work wait or fail?
Queue-to-assignment	Queue, Task, Assignment, Resource	enter queue, prioritize, assign, accept, reassign, start	Is delay demand-side, dispatch-side, or resource-side?
Readiness synchronization	Encounter/Case, Bed/Bay, Staff Assignment, Task	ready, resource-ready, staff-ready, transfer/start	Which prerequisite arrives last?
Movement and occupancy	Encounter, Placement Request, Bed Stay, Location	request, assign, depart origin, arrive, occupy, leave	How long between decision, movement, and physical occupancy?
Handoff	Handoff, Encounter/Case, Sending Team, Receiving Team	initiate, acknowledge, accept, complete, clarify	Where is responsibility ambiguous or repeated?
Split/merge/batch	Parent Request, Child Task/Specimen/Item, Batch	split, pool, process batch, merge results	Does batching improve throughput or create tail delay?
Exception/rework	Task/Request, Exception, Actor/Resource	fail, reject, correct, repeat, override, cancel	What creates loops and avoidable touches?
Capability matching	Request, Resource, Capability, Assignment	assess, shortlist, reject, assign, override	Are hard constraints or preference rules driving delay?
Intervention-to-outcome	Signal, Recommendation, Action, Intervention, Metric	detect, propose, approve, enact, measure, sustain	Did the change happen, and did the process shift?
Data correction	Source Record, Event, DQ Finding, Correction	ingest, validate, quarantine, correct, replay	Can the analysis be trusted and reproduced?

5. Analytical model families available over OCEL

Model family	Purpose	Hospital use	Maturity/caution
Object-centric lifecycle explorer	Inspect events/objects and per-type lifecycles	validate extraction and understand object histories	foundational; not an improvement result by itself
OC-DFG	Show frequent directly-follows relationships per object type	first readable map for ED, beds, transport, OR	strong exploration value; weak formal semantics
Object-centric Petri net (OCPN)	Represent concurrency, synchronization, variable cardinality	formal performance and conformance	higher complexity; model-quality validation required
Object-centric BPMN	Communicate a multi-object normative or discovered model	stakeholder review and redesign	useful presentation; tool support less uniform

Model family	Purpose	Hospital use	Maturity/caution
Object-centric behavioral constraints	Express declarative obligations/prohibitions	flexible clinical and safety pathways	often better than rigid sequences for variable care
Object-centric local process models	Find recurrent high-value fragments rather than a full spaghetti map	recurring discharge/transport/rework patterns	promising research area; newer technique
Variant/execution model	Compare coherent related-object executions	compare units, shifts, cohorts, pre/post periods	requires an explicit execution/case construction rule
OPerA performance model	Compute waiting, service, sojourn, flow, synchronization, pooling, and lagging	patient waits for bed; OR cases pool for PACU	formal OCPN replay; sensitive to model/log conformance
Object-centric conformance	Compare log behavior with a normative model	checklist, bundle, authorization, handoff, policy adherence	deviations require exception logic and clinical review
Organizational/resource model	Analyze roles, handoffs, workload, and collaboration	task switching, reassignment, coverage, handoff networks	worker surveillance and fairness risks
Comparative process cube	Slice by facility, unit, shift, service, cohort, or period	pre/post intervention and cross-site comparison	control for coding and source differences
Predictive process monitoring	Predict next event, remaining time, outcome, or workload	impending boarding breach, likely missed discharge	object-centric evidence/tooling still developing; validate locally
Object-centric simulation/digital twin	Generate what-if scenarios from discovered behavior	staffing, scheduling, beds, transport, PACU	discovery does not automatically produce a valid causal simulator
Intervention/action model	Close the loop from signal to governed action	route bottleneck to owner and measure response	keep human authority, audit, and balancing measures

OC-PM supports exploration, filtering, OC-DFG discovery, annotations, and some conformance metrics. OPerA defines formal object-centric performance measures. Object-centric local process models target recurring fragments. Object-centric simulation is promising for digital twins but remains methodologically demanding. [OC-PM](#), [OPerA](#), [Object-Centric Local Process Models](#), and [object-centric simulation](#).

6. Comprehensive hospital OCEL domain-model catalog

The catalog is intended to be exhaustive at the operational-domain level, not a command to implement every row. “Core interaction” names the minimum object collision that makes an object-centric model valuable.

6.1 Access and demand

ID	Candidate model	Core object-centric interaction	Primary improvement question	Evidence	Priority
A1	Referral-to-consult	Referral + Patient/Episode + Appointment + Service + Facility	Where do referrals wait, expire, redirect, or duplicate?	H+A	P2
A2	Elective waitlist and open-slot fill	Waitlist Entry + Appointment Slot + Patient + Service + Provider/Room	Why does capacity go unused while patients wait?	H+S+A	P2
A3	Pre-registration and eligibility	Encounter/Appointment + Coverage + Verification Task + Staff Assignment	Which incomplete steps create check-in delay or downstream denial?	A	P3
A4	Prior authorization / financial clearance	Service Request + Authorization + Payer + Task + Appointment/Case	Is delay caused by missing documentation, payer response, or internal rework?	S+A	P2
A5	Direct-admission intake	Admission Request + Encounter + Accepting Service + Bed/Unit + Transport	How long from acceptance to an operational destination?	S+A	P1
A6	ED end-to-end journey	Encounter + Patient + Orders + Results + Staff Assignments + Locations	Which variants and shared resources drive ED length of stay?	H	P1
A7	Ambulance offload	EMS Run + Encounter + Ambulance + ED Bay + Staff Assignment	Which object is unavailable when offload stalls?	S+A	P2
A8	ED admission and boarding	Encounter + Admit Decision + Placement Request + Bed Stay + Unit	Is boarding waiting on decision, bed, EVS, staffing, transport, or capability?	H+S+A	P1
A9	Transfer-center intake and acceptance	Transfer Request + Sending Facility + Receiving Facility + Service + Bed + Transport	Where do requests decline, queue, or wait after acceptance?	S+A	P1
A10	Scheduled arrival/no-show	Appointment + Patient + Service + Slot + Reminder/Offer	Which access processes lead to no-show, cancellation, or refill?	H+A	P2

ED process mining has identified examination/treatment loops associated with longer episodes, and hospital-wide reviews support system-level rather than ED-only flow analysis. [ED performance case study](#) and [IHI hospital-wide patient flow](#).

6.2 Capacity, placement, inpatient flow, and discharge

ID	Candidate model	Core object-centric interaction	Primary improvement question	Evidence	Priority
B1	Bed request and capability matching	Placement Request + Encounter + Bed + Unit + Capability + Decision	Which hard constraint or policy prevents timely assignment?	S+A	P1

ID	Candidate model	Core object-centric interaction	Primary improvement question	Evidence	Priority
B2	Assignment-to-physical occupancy	Placement Request + Bed Stay + Encounter + Origin + Destination + Transport	How much delay occurs after assignment but before actual occupancy?	S+A	P1
B3	Bed vacation and turnover	Bed Stay + Bed + EVS Task + Staff Assignment + Next Placement	Which lifecycle causes dirty-to-ready and ready-to-occupied delay?	S+A	P1
B4	Intra-hospital transfer	Transfer Request + Encounter + Origin Unit + Destination Unit + Bed Stay	Where does level-of-care progression stall?	H+S+A	P1
B5	ICU downgrade / stepdown	Encounter + Downgrade Decision + Placement Request + Bed/Unit + Staff Assignment	Is delay clinical, placement, staffing, or transport related?	S+A	P1
B6	Observation-to-inpatient/outpatient status	Encounter + Status Review + Authorization + Decision + Bed Stay	Which review steps cause status delay, rework, or compliance risk?	S+A	P2
B7	Discharge readiness and barriers	Encounter + Readiness State + Barrier + Owner + Service Request + Action	Which unresolved dependency creates medically-ready delay?	H+S+A	P1
B8	Post-acute placement	Encounter + Referral + Post-Acute Facility + Authorization + Transport	Is delay availability, acceptance, payer, document, or transport driven?	S+A	P1
B9	Discharge medication and departure	Encounter + Medication Order + Pharmacy Work + Education/Handoff + Transport	What prevents a ready patient from physically leaving?	H+S+A	P1
B10	RTDC forecast-plan-reconcile	Unit + Capacity Forecast + Demand Forecast + Huddle + Action + Outcome	Do forecast gaps become owned actions, and are they effective?	S+A	P1
B11	House huddle and escalation	Huddle + Barrier/Signal + Action + Owner + Unit/Service	Which issues recur because actions are late, unowned, or ineffective?	S+A	P1
B12	Surge, diversion, and incident command	Facility + Capacity State + Alert + Action + External Facility/EMS	Which actions stabilize flow under surge, and at what safety cost?	S+A	P2

The IHI framework explicitly treats the hospital as an interconnected system and recommends demand shaping, capacity matching, redesign, analytics, and a learning system. Health-information-system reviews show patient tracking, bed management, discharge management, order entry, and reporting as recurring flow interventions, while outcomes are mixed enough to demand local evaluation. [IHI](#) and [patient-flow HIS systematic review](#).

6.3 Perioperative and procedural operations

ID	Candidate model	Core object-centric interaction	Primary improvement question	Evidence	Priority
C1	Surgical booking and readiness	OR Case + Patient/Encounter + Appointment + Required Test/Clearance + Authorization	Which prerequisite makes a case non-ready or cancelled?	H+D+A	P1
C2	Block allocation, release, and fill	Block + OR Suite + Service + Case + Staff Assignment	Is access limited by allocation, late release, fill, or staffing?	H+A	P2
C3	First-case on-time start	OR Case + OR Suite + Patient + Team Assignment + Equipment/Tray	Which object is last ready at scheduled start?	H+A	P1

ID	Candidate model	Core object-centric interaction	Primary improvement question	Evidence	Priority
C4	Day-of-surgery flow	OR Case + Pre-op Bay + OR Suite + PACU Bay + Transport	Where do patient, room, team, and equipment lifecycles desynchronize?	H+D+A	P1
C5	Intraoperative phase and handoff	OR Case + Team + OR Suite + Handoff + Device/Tray	Which phase or handoff creates delay, rework, or risk?	H+D+A	P1
C6	Surgical-safety checklist conformance	OR Case + Checklist Instance + Team + Reference Model	Which required step is missing, late, repeated, or exception-approved?	H+S+A	P1
C7	OR room turnover	OR Suite + Prior Case + EVS Task + Next Case + Staff/Equipment	Is turnover waiting on cleaning, setup, staff, equipment, or patient?	H+A	P1
C8	PACU and downstream capacity	OR Case + PACU Bay + Staff Assignment + Inpatient/ICU Bed + Transport	Do cases pool because recovery, staffing, transport, or destination is late?	D+H+A	P1
C9	Cancellation and rework	OR Case + Cancellation/Exception + Prerequisite + Slot + Replacement Case	Which upstream failures waste capacity and how often is it recovered?	D+H+A	P1
C10	Sterile processing / instrument trays	Case + Tray/Set + Sterilization Cycle + Sterile Processor + Transport	Do missing, late, incomplete, or reprocessed trays delay cases?	S+A	P2
C11	Cath/EP/IR/endoscopy flow	Procedure Case + Room + Team + Device/Supply + Recovery Bay + Bed	Which shared resource governs throughput?	H+A	P2
C12	Procedure-to-inpatient dependency	Procedure Case + Encounter + PACU/Recovery + Bed/Unit + Transport	How much elective work is delayed by downstream hospital capacity?	D+H+A	P1

Perioperative process mining has used EHR audit logs, observation, interviews, and RTLS data to study workflows and staff interactions. The 2026 OCPM study found major delay outside the intraoperative core, particularly preoperative and PACU/discharge transitions. [Mayo pre-op study](#), [OR dashboard/RTLS study](#), and [Celik OCPM study](#).

6.4 Diagnostics, consults, therapies, and medication

ID	Candidate model	Core object-centric interaction	Primary improvement question	Evidence	Priority
D1	Lab order-to-result	Service Request + Specimen + Test + Analyzer + Result + Encounter	Is turnaround driven by collection, transport, accession, queue, analyzer, validation, or notification?	H+S+A	P1
D2	Specimen split/merge/recollection	Parent Specimen + Aliquot + Tests + Analyzer + Recollection Request	Where do batching, add-ons, insufficient samples, or recollection create delay?	H+A	P2
D3	Critical-result communication	Result + Encounter + Responsible Team + Communication Task + Acknowledgment	Are critical results acknowledged and acted upon reliably?	S+A	P2
D4	Blood product lifecycle	Order + Blood Product Unit + Patient/Encounter + Cooler/Location + Transfusion	Where are units allocated, delayed, returned, expired, or wasted?	H+S+A	P2
D5	Imaging order-to-final report	Service Request + Appointment/Queue + Imaging Study + Scanner + Radiologist Assignment	Is delay protocol, scheduling, transport, acquisition, interpretation, or signoff?	H+S+A	P1
D6	Imaging critical finding / result communication	Imaging Study + Result + Encounter + Communication Task + Responsible Team	Are urgent findings acknowledged and resolved?	S+A	P2

ID	Candidate model	Core object-centric interaction	Primary improvement question	Evidence	Priority
D7	Pathology specimen-to-diagnosis	Specimen + Accession/Case + Block/Slide + Test + Pathologist Assignment + Report	Which processing, batching, referral, or review step drives turnaround?	H+A	P2
D8	Specialty consult	Consult Request + Encounter + Consulting Service + Assignment + Recommendation	Is delay in request quality, acceptance, staffing, evaluation, or decision?	H+S+A	P1
D9	PT/OT/SLP therapy clearance	Therapy Request + Encounter + Therapist Assignment + Session + Disposition Barrier	Which evaluation or equipment dependency blocks progression?	H+A	P1
D10	Respiratory therapy / device workflow	Order + Encounter + Device + Therapist Assignment + Treatment	Are treatments delayed by device, staffing, location, or order changes?	A	P2
D11	Medication order-to-administration	Medication Order + Pharmacy Work + Medication/Dose + Encounter + Nurse Assignment	Where do verification, preparation, delivery, or administration delays occur?	H+S+A	P2
D12	Discharge medication / medication reconciliation	Medication Order + Pharmacy Work + Encounter + Education Task + Caregiver/Transport	Which object prevents safe, timely discharge?	H+S+A	P1
D13	High-cost/specialty medication authorization	Medication Order + Authorization + Pharmacy Work + Payer + Encounter	Where does clinical or payer documentation create delay?	S+A	P2
D14	Nutrition/dietary fulfillment	Nutrition Order + Meal/Feed Task + Encounter + Unit + Delivery	Which order changes, production, or delivery issues cause misses?	A	P3

Clinical chemistry process mining at two university hospitals modeled sample flow and measured turnaround, timeliness, work in process, workload, and machine downtime. A 2025 hospital pathway study found laboratory and radiology services to be important shared-resource bottlenecks.

[Laboratory process-mining study](#) and [hospital pathway study](#).

6.5 Logistics, workforce, assets, and facilities

ID	Candidate model	Core object-centric interaction	Primary improvement question	Evidence	Priority
E1	Internal patient transport	Transport Job + Encounter + Origin + Destination + Assignment + Vehicle	Is delay request quality, dispatch, resource arrival, patient readiness, or destination readiness?	H+S+A	P1
E2	External discharge/medical transport	Transport Job + Encounter + Vendor/Vehicle + Authorization + Destination	Which booking, payer, vendor, or readiness step delays departure?	S+A	P1
E3	EVS task dispatch and execution	EVS Task + Room/Bed + Staff Assignment + Supply/Equipment + Placement	Is dirty-to-ready delay dispatch, travel, cleaning, inspection, or blockage?	S+A	P1
E4	Staffing plan-to-deployment	Shift + Staff Assignment + Unit + Role/Capability + Demand State	Are scheduled staff present, deployable, skill-matched, and aligned to demand?	S+A	P1
E5	Workload and task distribution	Task + Staff Assignment + Encounter/Location + Queue	Are workload, reassignments, interruptions, and overdue work uneven?	H+A	P2
E6	Handoff/collaboration network	Handoff + Sending Assignment/Team + Receiving Assignment/Team + Encounter	Where do acknowledgments, clarification, or rework cluster?	H+A	P2

ID	Candidate model	Core object-centric interaction	Primary improvement question	Evidence	Priority
E7	Equipment/asset request and use	Asset Request + Device/Asset + Encounter/Location + Transport/Maintenance Task	Is care waiting on locating, cleaning, charging, transport, or maintenance?	S+A	P2
E8	Supply replenishment and stockout	Supply Item/Lot + Location + Replenishment Task + Procedure/Encounter	Which stockouts or substitutions disrupt work?	S+A	P2
E9	Pharmacy/material delivery logistics	Delivery Job + Medication/Supply + Origin + Destination + Robot/Staff	Which route, batch, handoff, or destination creates delay?	A	P3
E10	Facilities maintenance	Work Order + Asset/Room + Technician Assignment + Part + Operational Impact	Which failure and repair paths remove clinical capacity?	A	P2
E11	Environmental isolation/terminal cleaning	Encounter + Room + Isolation Requirement + EVS Task + Infection-Control Clearance	How do enhanced cleaning and clearance affect capacity?	S+A	P2
E12	Waste/linen/material reverse logistics	Collection Task + Container/Item + Location + Staff/Vehicle	Where do accumulation, pickup, or return cycles disrupt operations?	A	P3

Patient-transport process mining has been used to identify bottlenecks and delay causes in a German hospital. Healthcare-logistics research is still fragmented; a 2026 systematic-review protocol explicitly scopes supply chains, intrahospital transport, inventory, equipment tracking, and medication distribution as distinct logistical process-mining domains. [Patient transport study](#) and [BMJ Open logistics review protocol](#).

6.6 Quality, safety, reliability, and clinical pathways

ID	Candidate model	Core object-centric interaction	Primary improvement question	Evidence	Priority
F1	Sepsis time-critical pathway	Pathway Instance + Encounter + Orders + Specimens/Results + Medications + Team	Which required step is missing, late, out of order, or exception-justified?	H+S+A	P2
F2	Stroke pathway	Pathway Instance + Encounter + Imaging + Result + Medication/Procedure + Team	Where do door-to-imaging/read/decision delays occur?	H+S+A	P2
F3	STEMI/trauma/other time-critical pathway	Pathway Instance + Encounter + Team + Procedure/Room + Transport	Which interdependent resource causes time-critical delay?	H+S+A	P2
F4	Boarding safety and reassessment	Boarded Encounter + Location + Staff Assignment + Medication/Task + Alert	Are reassessment, medication, monitoring, and escalation reliable while boarding?	S+A	P1
F5	Rapid response / escalation	Encounter + Signal + Alert + Team Activation + Intervention + Transfer	Is detection, activation, team arrival, intervention, or destination late?	S+A	P2
F6	Infection prevention bundle	Encounter + Device/Procedure + Staff/Team + Checklist + Observation/Result	Which bundle elements or device-duration processes deviate?	S+A	P2
F7	Isolation placement and exposure	Encounter + Condition/Requirement + Room/Bed + Contact/Staff Assignment + EVS	How long until appropriate placement and clearance?	S+A	P2
F8	Medication safety event	Medication Order + Pharmacy Work + Dose/Admin + Encounter + Incident	Which upstream path and handoff precede an error or near miss?	H+S+A	P2

ID	Candidate model	Core object-centric interaction	Primary improvement question	Evidence	Priority
F9	Falls/pressure injury prevention process	Encounter + Risk Assessment + Intervention Task + Staff Assignment + Incident	Were preventive processes completed and sustained?	S+A	P3
F10	Incident-to-CAPA	Incident + Investigation + Finding + Action + Owner + Metric	Do investigations produce timely, completed, effective corrective actions?	S+A	P2
F11	Readmission/cross-encounter transition	Patient + Prior Encounter + Follow-up + Medication/Handoff + New Encounter	Which transition patterns are associated with return, without claiming causality?	H+A	P2
F12	Equity in access and flow	Encounter/Request + Service/Facility + Operational Process + Governed Demographics	Do delays or variants differ across groups after missingness and small-cell controls?	S+A	P2
F13	Measure-evidence lineage	Metric Definition + Source Event/Result + Computation + Review + Published Measure	Can every reported measure be reproduced and versioned?	S+A	P1

Conformance checking can compare actual care with guidelines, but flexible pathways, justified exceptions, data latency, and documentation gaps make rigid sequence models risky. Declarative constraints and object-centric local patterns may be more appropriate for many clinical processes. A stroke case study demonstrated time-aware bottleneck analysis, while a rectal-cancer study applied process mining to guideline compliance. [Stroke process mining](#) and [rectal-cancer conformance](#).

6.7 Administrative, financial, enterprise, and network operations

ID	Candidate model	Core object-centric interaction	Primary improvement question	Evidence	Priority
G1	Utilization review / continued stay	Encounter + Review + Authorization + Payer + Documentation Task + Decision	Which documentation, review, or payer loop delays status/disposition?	S+A	P2
G2	Charge capture and coding	Encounter/Procedure + Charge Item + Documentation + Coder Task + Claim	Where are charges delayed, missing, corrected, or rejected?	A	P3
G3	Claim submission and denial	Claim + Encounter + Payer + Denial + Appeal Task + Payment	Which upstream process patterns generate rework and preventable denial?	A	P3
G4	Patient financial assistance/counseling	Encounter + Coverage + Application + Task + Decision	Where do incomplete applications or handoffs create delay?	A	P3
G5	Credentialing and privileging	Practitioner + Credential + Privilege + Review Task + Facility/Service	Which dependencies delay safe deployment of workforce capacity?	A	P3
G6	Hiring/onboarding/education compliance	Employee + Position + Credential/Training + Task + Assignment	Where does onboarding stall or require rework?	A	P3
G7	Procurement-to-pay	Requisition + Purchase Order + Item + Shipment + Invoice + Facility	Which supply dependencies threaten operations?	S+A	P3
G8	Multi-facility transfer/load balancing	Transfer Request + Sending/Receiving Facility + Capability + Bed + Transport	Which geography/capability edge limits network access?	S+A	P1
G9	Service-line longitudinal journey	Patient/Episode + Encounters + Facilities + Service Line + Referrals/Procedures	Where does the network lose continuity or create avoidable travel/wait?	H+A	P2

ID	Candidate model	Core object-centric interaction	Primary improvement question	Evidence	Priority
G10	Ambulatory-inpatient transition	Patient/Episode + Appointment + Encounter + Admission/Discharge + Handoff	Are follow-up and escalation pathways connected?	H+A	P2
G11	Data-feed and application reliability	Source System + Interface/Feed + Event Batch + DQ Finding + Incident + Metric	Which technology failure distorts operational truth or delays work?	S+A	P1
G12	Downtime and recovery	Downtime Incident + Application/Device + Department + Manual Workaround + Recovery Task	How do outages alter work, safety exposure, and recovery?	S+A	P2

These models are valid OCEL opportunities but have less healthcare-specific process-mining evidence than flow, perioperative, laboratory, and transport models. They should follow clinical-operational models unless a hospital's strategic pain makes one a funded exception.

6.8 Improvement and governance of the operating system itself

ID	Candidate model	Core object-centric interaction	Primary improvement question	Evidence	Priority
H1	Signal-to-alert lifecycle	Metric/Signal + Alert + Unit/Service + Owner	Are signals acknowledged, suppressed, escalated, and cleared appropriately?	S+A	P1
H2	Recommendation-to-action	Recommendation + Approval + Operational Action + Owner + Affected Objects	Which recommendations become timely work, and where do they fail?	S+A	P1
H3	Huddle action closure	Huddle + Action + Barrier + Owner + Unit/Encounter	Do huddles resolve constraints or repeatedly rediscover them?	S+A	P1
H4	PDSA/intervention lifecycle	Opportunity + PDSA + Intervention + Cohort/Scope + Metric + Outcome Review	Was the change implemented, sustained, adapted, or abandoned?	S+A	P1
H5	Reference-model governance	Policy/Guideline Version + Process Model + Approver + Conformance Run	Which standard version generated a deviation, and is it current?	S+A	P1
H6	Model-review and publication	Discovered Model + Evidence Window + Reviewer + Approval + Published View	Is a map valid, interpretable, and approved for use?	A	P1
H7	Data-quality issue-to-correction	DQ Finding + Source + Mapping/Rule + Owner + Correction + Replay	Are defects detected, repaired, and prevented?	S+A	P0
H8	Forecast/recommendation reconciliation	Forecast + Recommendation + Action + Actual Outcome + Review	Was a prediction accurate and was advice adopted?	S+A	P1

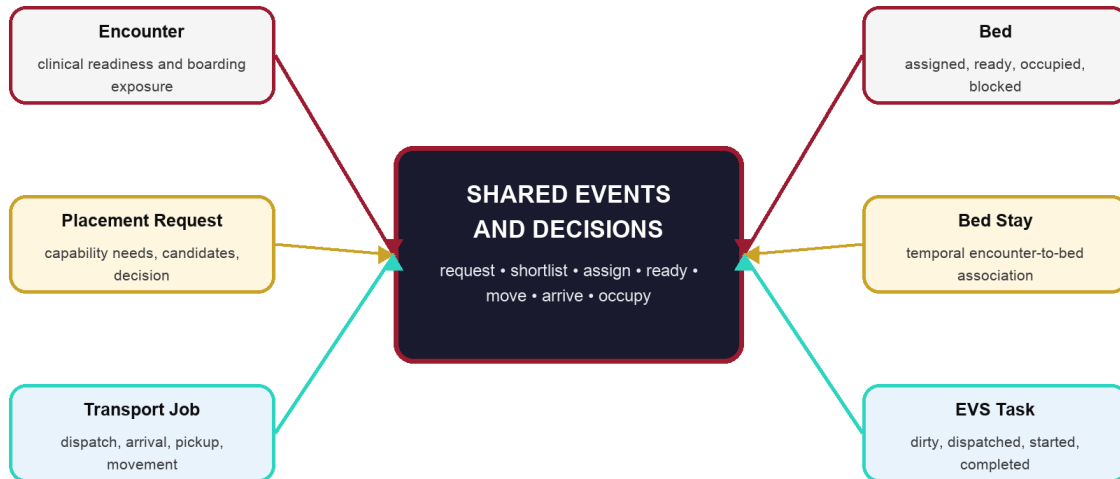
These models create the learning loop demanded by hospital QAPI. They also prevent the OCEL platform from becoming a passive dashboard that detects the same bottleneck every week.

7. Detailed designs for the P1 model portfolio

7.1 ED admission, boarding, placement, and physical occupancy

Purpose: distinguish the clinical decision to admit from request creation, assignment, bed readiness, physical movement, and actual occupancy.

The ED-to-bed process is a synchronization problem, not one patient trace



Object-centric performance asks which prerequisite arrived last and which object waited.

Figure 2. ED-to-bed flow is a multi-object synchronization problem. Temporal association objects prevent assignment and occupancy from being conflated.

Objects: Patient (optional), Encounter, Admit Decision, Placement Request, Placement Candidate/Decision, Bed Stay, Bed, Unit, Transport Job, EVS Task, Staff Assignment, Barrier.

Minimum events:

- admit-decision-recorded;
- placement-requested;
- candidate-rejected with reason;
- bed-assigned;
- bed-ready;
- transport-requested, transport-arrived, patient-picked-up;
- departed-origin;
- arrived-destination;
- physical-occupancy-started;
- placement-cancelled **OR** assignment-revoked.

Key relationships:

- Encounter has Placement Request;
- Placement Request targets capability/unit, not a bed until decided;

- Bed Stay binds Encounter and Bed for a validity interval;
- Transport Job moves Encounter from origin to destination;
- Staff Assignment performed decision or movement event.

Metrics: decision-to-request, request-to-assignment, assignment-to-bed-ready, bed-ready-to-origin-departure, origin-departure-to-destination-arrival, assignment-to-physical-occupancy, boarding exposure, assignment reversals, candidate rejection reasons, capability mismatch, off-service placement, p50/p90 and tail exposure.

Analyses: OC-DFG for exploration; OCPN/OPerA for synchronization among encounter, bed, transport, and staff; conformance for local placement policy; comparative cube by unit, shift, service, and capability.

Critical balancing measures: placement-related safety events, transfers after placement, nursing workload, isolation/cohorting reliability, subgroup differences, and overrides.

7.2 Bed vacation and EVS turnover

Purpose: measure the real bed lifecycle and determine whether turnover delay is driven by upstream departure, notification, dispatch, cleaning, inspection, blockage, or next-placement coordination.

Objects: Bed, Bed Stay, Encounter, EVS Task, EVS Assignment, Staff Assignment, Room, Isolation Requirement, Maintenance Work Order, Next Placement Request.

Minimum events: patient-physically-left, bed-vacated, dirty-status-set, cleaning-requested, task-dispatched, cleaning-started, cleaning-completed, inspection-completed, bed-ready, bed-blocked, maintenance-started/completed, next-occupancy-started.

Do not substitute: discharge order for patient departure; bed assignment for occupancy; completed discharge encounter for bed vacancy; EVS dispatch for cleaning start.

Metrics: vacated-to-dirty notification, dirty-to-dispatch, dispatch-to-start, service time, completion-to-ready, ready-to-next-assignment, ready-to-next-occupancy, re-clean rate, blocked time, priority changes, staff travel time when available.

Object-centric value: OPerA can distinguish patient/bed synchronization, EVS task pooling, and ready-bed lagging. A patient-only model cannot measure a ready bed waiting for the next placement.

7.3 Discharge readiness, barriers, and actual departure

Purpose: identify which patient, system, payer, post-acute, pharmacy, transport, caregiver, or documentation dependency prevents safe departure.

Objects: Encounter, Discharge Plan, Readiness Assessment, Barrier, Barrier Ownership, Service Request, Authorization, Post-Acute Referral, Medication Order, Pharmacy Work, Transport Job, Follow-up Appointment, Operational Action.

Minimum events: expected-discharge-date-set/revised, clinical-ready, disposition-ready, barrier-opened, owner-assigned/acknowledged, evidence-received, barrier-resolved, discharge-order-entered, medications-ready, education-completed, transport-ready, patient-physically-departed, bed-vacated.

Modeling rule: overlapping barriers must be interval-aware. Do not add each barrier's duration and call the sum "avoidable days." Use exposure intervals, primary-constraint logic, or multi-state attribution with explicit uncertainty.

Metrics: medically-ready-to-departure, barrier acknowledgment and resolution, time unowned, recurrence, authorization wait, referral acceptance wait, medication-ready delay, transport wait, order-to-departure, follow-up completion, 7/30-day return as balancing outcomes.

7.4 RTDC forecast, huddle, action, and reconciliation

Purpose: model RTDC as a management process, not a forecast dashboard.

Objects: Unit, Capacity Forecast, Demand Forecast, Huddle, Plan, Operational Action, Barrier, Owner/Assignment, Actual Capacity/Demand Observation, Reconciliation.

Minimum events: forecast-published, forecast-revised, gap-detected, huddle-opened, plan-approved, action-assigned, action-accepted, action-started, action-completed/failed/overridden, actual-observed, reconciled, improvement-referral-created.

Metrics: forecast error/bias/coverage, plan coverage, action acceptance and completion, overdue exposure, escalation, recurrence, gap closure, forecast-to-action latency, action-to-outcome latency.

Object-centric value: the model joins one huddle and its actions to many units, encounters, beds, barriers, and owners without duplicating the huddle as if each patient had an independent management cycle.

7.5 Transfer center and multi-facility access

Purpose: expose the full request-to-acceptance-to-arrival network, including declined, rerouted, and abandoned demand.

Objects: Transfer Request, Patient/Encounter, Sending Facility, Receiving Facility, Service, Capability, Accepting Clinician/Assignment, Bed/Placement Request, Authorization, Transport Job, Document Package.

Minimum events: request-received, clinical-info-requested/received, capability-screened, clinician-contacted, accepted/declined, bed-requested/assigned, authorization-started/completed, transport-booked/arrived, departed-sending, arrived-receiving, request-cancelled/rerouted.

Metrics: request-to-first-review, clinical decision, acceptance, bed assignment, transport, and arrival; decline reasons; repeated contacts; lost demand; capability mismatch; accepted-but-never-arrived; network imbalance.

7.6 Perioperative readiness, OR, PACU, and downstream capacity

Purpose: prevent local OR optimization from shifting delay and risk into pre-op, PACU, ICU, or inpatient units.

Objects: OR Case, Encounter, OR Suite, Pre-op Bay, PACU Stay/Bay, Bed/Unit, Team Assignment, Instrument Tray, Device, Transport Job, Readiness Requirement, Cancellation/Exception.

Minimum events: booked, readiness-required/completed, case-ready, patient-ready, room-ready, team-ready, tray-ready, wheels-in, procedure-start/end, wheels-out, PACU-arrival, PACU-ready-for-discharge, destination-ready, PACU-departure, turnover-start/complete, cancelled.

Metrics: booking-to-procedure, first-case start components, readiness completeness, case-duration error, turnover components, cancellations, PACU recovery vs hold, PACU pooling, after-hours utilization, downstream-capacity cancellation.

Critical design: model PACU as its own location/resource lifecycle. Do not mark the OR suite occupied during recovery unless the patient is physically still in the room.

7.7 Transport request-to-dropoff

Purpose: separate request, acceptance, assignment, dispatch, transporter arrival, patient readiness, pickup, transit, handoff, and completion.

Objects: Transport Job, Encounter, Origin, Destination, Transport Assignment, Transporter/Vehicle, Equipment, Barrier.

Minimum events: requested, accepted, assigned, dispatched, arrived-origin, patient-ready, pickup-started, departed-origin, arrived-destination, handoff-acknowledged, completed, cancelled, escalated, reassigned.

Metrics: request-to-acceptance, dispatch, arrival, pickup, transit, handoff, completion; readiness mismatch; cancellations; reassignments; priority escalation; missed diagnostic/procedure slot; idle resource time.

Critical design: dispatch and assignment are not pickup. If pickup is not recorded, report a dispatch proxy rather than renaming it.

7.8 Lab, imaging, and consult shared-service models

These three models share a request-to-fulfillment pattern but need different object granularity.

Domain	Objects that must remain distinct	Most important waits
Laboratory	Service Request, Specimen, Aliquot, Test, Analyzer, Result	order-to-collection, transport, accession, analyzer queue, validation, notification
Imaging	Service Request, Appointment/Queue, Imaging Study, Scanner, Transport, Radiologist Assignment, Report	protocol, schedule, patient ready/transport, scanner, read, final signoff
Consult	Consult Request, Encounter, Consulting Service, Assignment, Evaluation, Recommendation	request quality, acceptance, assignment, bedside evaluation, decision/recommendation

For all three, record cancellation, insufficient information, add-on work, repeat/recollection, priority change, and critical-result acknowledgment. These are often the rework pathways that simple order/result timestamps hide.

7.9 Staffing plan-to-deployment and workload

Purpose: join demand and work to deployable skill, not raw headcount.

Objects: Shift, Staff Assignment, Role/Credential, Unit, Task/Queue, Patient Assignment where governed, Break/Coverage Assignment, Surge Request.

Minimum events: scheduled, confirmed, clocked-in, deployed, reassigned, coverage-requested/accepted, task-assigned/accepted/completed, break-start/end, clocked-out.

Metrics: scheduled vs present vs deployable vs skill-matched capacity; workload and overdue work; reassignment; interruption; time outside home unit; coverage response; overtime and fatigue balancing measures.

Governance: use assignment-level pseudonymous data by default. Do not turn process mining into individual productivity ranking without labor, legal, privacy, fairness, and clinical-governance review.

7.10 Improvement lifecycle and intervention attribution

Purpose: prove whether detected problems became real, completed, sustained improvement work.

Objects: Opportunity, Signal/Alert, Recommendation, Approval, Operational Action, PDSA/Intervention, Owner, Metric Definition, Outcome Observation, Balancing Measure, Reference Model Version.

Minimum events: opportunity-detected, recommendation-drafted, approved/rejected, action-assigned, started, completed/failed/overridden, intervention-started/ended, measure-observed, reviewed, adopted/adapted/abandoned, sustainment-check.

Metrics: signal-to-owner, approval, start, and completion; workflow fidelity; intervention exposure; recurrence; sustained shift; balancing-measure change; model/version provenance.

Causal boundary: this model can document temporal association and implementation fidelity. Causal claims require an appropriate evaluation design.

8. Object and event naming conventions

8.1 Object naming

- Use singular nouns: Encounter, Placement Request, Bed Stay.
- Do not combine concepts: avoid PDSA / Intervention; choose distinct types if lifecycles differ.
- Make association objects explicit: Staff Assignment, not a timeless Staff assigned-to Unit edge.
- Keep clinical identity separate from operational work identity.
- Version the catalog and attributes; additive growth alone is insufficient when meaning changes.

8.2 Event naming

- Use completed past-tense operational facts: bed-assigned, patient-arrived, cleaning-started.
- Do not use vague verbs such as update, process, status-change, or complete without a domain noun.
- Keep intent, execution, and documentation distinct.
- Use one event for one real activity execution. A database-row snapshot is not automatically an event.
- Preserve corrections and supersession. Do not silently rewrite history used for prior decisions.

8.3 Qualifier vocabulary

Use a controlled, versioned set. Recommended E2O qualifiers include:

subject, patient, encounter, request, task, actor, owner, approver, performer, resource, origin, destination, location, result, trigger, target, predecessor, successor, evidence, affected, balancing-measure.

Recommended stable O2O qualifiers include:

of, in, part-of, contains, requested-by, fulfills, result-of, instantiates, replaces, triggered-by, measures, governed-by.

Avoid using O2O for temporal occupancy, assignment, ownership, or device-use relationships unless the validity is modeled elsewhere.

9. Data-source architecture

9.1 Source classes

Source class	Likely objects/events	Key limitations
HL7 v2 ADT / EHR encounter data	arrivals, admissions, transfers, discharges, locations	administrative event may lag physical movement
FHIR APIs/events	Encounter, Task, ServiceRequest, Procedure, Observation, Medication, Location	availability and resource support vary by vendor/site
EHR audit logs	user interactions, chart access, orders, documentation	records system interaction, not always care execution
Bed/flow/transfer-center systems	placement requests, decisions, states, transfer calls	local status meanings and manual workarounds
RTLS / badge / device telemetry	physical arrival/departure, asset movement	coverage, badge sharing, detection zones, privacy
OR/anesthesia/PACU systems	case milestones, room/bay phases, staffing	timestamp definitions vary; scheduled vs actual confusion
LIS/middleware/analyzers	specimen/test/analyzer events	multiple identifiers, split/merge, instrument clock drift
RIS/PACS/dictation	imaging work and reporting	order/study/accession identifiers require reconciliation
Pharmacy/BCMA	order, verify, prepare, dispense, administer	clinical exceptions and late documentation
Workforce/timekeeping	shifts, presence, role, assignment	scheduled is not deployable; sensitive employee data
Transport/EVS/CMMS	job queues, assignments, status changes	coarse status and incomplete start/arrival times
ERP/supply chain	item, requisition, purchase, shipment, inventory	separate master-data and facility-location conventions
Payer/UM/RCM	authorization, review, claim, denial, appeal	long latency and external-party opacity
Quality/safety/PDSA systems	incidents, actions, measures, interventions	retrospective entry and inconsistent taxonomies

9.2 FHIR and OMOP roles

FHIR is most useful as an interoperability and workflow-semantic source. Its Request/Event/Task relationships help define object lifecycles, but a FHIR repository is not automatically a reliable event log.

OMOP is valuable for retrospective normalization and cross-site clinical research. OMOP event tables link clinical occurrences to visits, and the 2024 OMOP-CDM-to-OCEL study shows a feasible transformation. OMOP generally does not preserve every operational queue, dispatch, ownership, physical movement, or workflow timestamp needed for real-time hospital operations. [OMOP CDM](#).

The canonical OCEL should therefore integrate operational sources directly and use FHIR/OMOP as mappings—not force every operational event through either model.

10. Data quality and validation gates

10.1 Required checks before any process result is published

- event uniqueness and deterministic source identity;
- referential integrity for every E2O/O2O relation;
- object clone and object conflation tests;
- missing and false event-object relationships;
- missing and false object-object relationships;
- unexpected object-attribute changes;
- timestamp ordering, clock drift, future timestamps, and negative durations;
- documentation-proxy labeling;
- late-arriving and corrected-event rates;
- source-to-OCEL reconciliation by window, domain, and activity;
- lifecycle completeness by object type;
- qualifier cardinality and controlled-vocabulary validation;
- model fitness, precision, simplicity, and generalization where applicable;
- manual trace review with domain experts;
- reproducibility from source snapshot and mapping version.

A 2026 object-centric data-quality study identifies eight recurring imperfection patterns: object clones, different objects sharing one ID, inconsistent/mutating object attributes, missing/incorrect O2O relations, and missing/incorrect E2O relations. All eight appeared in real data evaluated by the authors. [Object-centric event-data imperfection patterns](#).

10.2 Model-specific acceptance

Model class	Minimum acceptance gate
OC-DFG	frequency reconciliation, stable filtering, expert face-validity
OCPN	fitness, precision, simplicity, concurrency and cardinality review
Performance	timestamp provenance, lifecycle completeness, formal metric definition, tail validation
Conformance	approved reference/version, exceptions, latency tolerance, false-positive review
Prediction	temporal validation, calibration, subgroup performance, operational utility, drift
Simulation	input/distribution validation, face validity, backtesting, sensitivity, scenario assumptions
Intervention	pre-specified outcome/process/balancing measures and implementation-fidelity checks

11. Privacy, security, and workforce governance

Tokenization or hashing does not by itself establish HIPAA de-identification. HHS recognizes Safe Harbor and Expert Determination methods and emphasizes assessment of linkage and re-identification risk. Exact event times, rare pathways, facilities, roles, and longitudinal patterns can be identifying even after names and MRNs are removed. [HHS de-identification guidance](#).

Minimum governance:

- define whether each OCEL use is treatment/payment/operations, QI, research, or another purpose;
- keep minimum necessary fields and scoped access;

- separate operational identity resolution from analysis tokens;
- prefer keyed HMAC/token vaults over unsalted stable hashes for identifiers;
- prohibit free text by default;
- suppress small cells and rare-path exports;
- control exact timestamps and geography in external exports;
- audit every query and export;
- use retention/archival rules by purpose;
- review worker-level analytics with workforce, labor, legal, ethics, and clinical governance;
- prevent individual ranking from proxy measures such as click counts or task completion without context;
- require human approval before process-mining output changes patient care or workforce assignment.

Privacy-preserving process-mining research notes that both attribute values and atypical process behavior can identify individuals. [Privacy-Preserving Process Mining in Healthcare](#).

12. Zephyrus current-state assessment

Reviewed read-only at /Users/sudoshi/Github/Zephyrus, main, commit e6dbda4 on 2026-07-09.

12.1 What exists

- `database/migrations/2026_07_04_000200_create_ocel_schema.php` defines catalog, event, object, E2O, O2O, and object-change tables.
- `app/Domain/Ocel/OcelCatalog.php:31-47` declares 12 object types; seven are marked emitted: Patient, Encounter, Bed, Unit, OR Case, OR Suite, and Transport Job.
- Order, EVS Task, Staff Assignment, Alert, and PDSA/Intervention are declared but not emitted (`OcelCatalog.php:41-46`).
- `app/Domain/Ocel/EmissionMap.php` projects flow, perioperative milestone/timing, and transport events.
- `arena/app/discovery.py:45-95` builds an OC-DFG-like union by flattening once per object type and discovering a DFG per type.
- `arena/app/performance.py:34-119` computes per-object handoff gaps and a heuristic multi-object wait view.
- `arena/app/conformance.py:23-92` runs batch rule checks by one configured case object type.
- Arena routes/UI, cached performance/conformance signals, and governed copilot code exist behind flags, so the older Part X status table describing X1-X4 as pending is stale as an implementation inventory.

12.2 Material semantic gaps

Gap	Current evidence	Why it matters	Required correction
Timeless occupancy relation	<code>EmissionMap.php:99-108</code> emits Encounter occupies Bed as O2O; O2O table has no validity time	a transferred encounter can appear to occupy every historical bed simultaneously	introduce Bed Stay/Placement association object and begin/end events
Assignment equals occupancy	<code>EmissionMap.php:110-121</code> sets bed occupied for assign-bed	assignment is not physical presence; boarding measurement becomes biased	split assigned, ready, departed-origin, arrived, occupied
Discharge equals vacancy	same block marks bed vacated on generic discharge	encounter discharge/order/physical departure/bed vacancy differ	source physical departure/vacancy explicitly

Gap	Current evidence	Why it matters	Required correction
Dispatch equals pickup	EmissionMap.php:312-315 maps dispatched_at or assigned_at to transport-pickup	request-to-pickup metrics become semantically false	preserve assigned, dispatched, arrived-origin, pickup separately; label proxies
Recovery occupies OR suite	EmissionMap.php:242-251 defaults non-procedure/non-turnover phase to suite occupied	PACU recovery may happen outside the OR	model PACU Stay/Bay and close OR occupancy at wheels-out
Object-change idempotency is not enforced	OcelProjector.php:275-278 uses insertOrIgnore; migration only indexes (object_id, changed_at) at lines 124-133	reruns can append duplicate state changes because no natural conflict exists	add unique (object_id, attr, changed_at, source_ref/version) or deterministic change ID
Empty object attribute schemas	OcelProjector.php:291-303 writes {} for every attrs_schema	catalog version does not validate object meaning	publish versioned JSON schemas and migration compatibility rules
Heuristic performance labeled OPerA-style	performance.py uses adjacent event deltas and time since each object's prior event	this is useful but not formal OPerA synchronization/pooling/lagging from OCPN replay	label metrics as heuristic; add OCPN replay before claiming OPerA
Batch rule conformance	conformance.py:34-92 collects first timestamps per activity and calls a custom evaluator	it is not object-centric Petri-net alignment or streaming prefix conformance	retain as explicit rule engine; add formal model/exception/version support if needed
Weak de-identification claim	EmissionMap::hashRef() uses truncated unsalted SHA-256	deterministic hashing and exact event patterns may remain linkable; "PHI-safe" is not established by hashing alone	use a governed token service/HMAC and formal privacy assessment
Mixed activity naming	catalog mixes kebab case, snake case, title case, and generic activities	discovery fragments semantically identical work and complicates cross-site comparison	version a canonical activity ontology and source mappings

12.3 What the current Arena can and cannot claim

Capability	Defensible current description	Claim to avoid
Discovery	per-object-type DFGs combined into an object-tagged view	full formal discovery of an object-centric Petri net
Performance	adjacent lifecycle gaps and heuristic wait-at-multi-object-event summaries	complete OPerA synchronization, pooling, and lagging analysis
Conformance	batch, pathway-specific rules over one case object type	general object-centric conformance or online prefix alignment
Data coverage	flow events, selected perioperative events, and transport phases	hospital-wide OCEL or complete ED/OR/transport physical truth
Privacy	identifiers are transformed and free text is excluded in the emission map	HIPAA de-identification or universally PHI-free data without formal review

This distinction does not diminish the foundation. It makes the next build phase measurable and prevents premature outcome or safety claims.

13. Recommended Zephyrus target architecture

13.1 Canonical layers

1. **Raw immutable source envelope** — source payload/reference, source timestamps, ingestion timestamp, correction/supersession, facility, interface version.
2. **Canonical operational events** — governed activity ontology and identity resolution, still source-faithful.
3. **OCEL projection** — event/object types, qualified relations, association objects, time-varying attributes, versioned schemas.
4. **Validated domain views** — ED/placement, discharge, EVS, perioperative, transport, etc.
5. **Model registry** — scope, object types, filters, algorithm/library version, evidence window, fitness/precision, owner, approval, expiration.
6. **Metric registry** — definition, denominator, time semantics, source coverage, thresholds, balancing measures.
7. **Governed action plane** — recommendation, approval, assignment, execution, override, outcome review.

13.2 Minimum schema additions

- deterministic ID and source provenance on every object change;
- natural uniqueness for object changes;
- mapping version and source-event version;
- **event** `documented_at`, `ingested_at`, `corrected_at`, `time_basis`, and `precision` attributes;
- object-type and event-type schema validation;
- catalog deprecation/replacement semantics;
- association-object pattern for every temporal many-to-many relation;
- correction/supersession events rather than silent updates;
- facility/tenant scope on events, objects, and exports;
- row-level access/retention/export policy metadata;
- domain-view registry and reproducible extraction parameters.

13.3 First bounded views

View	Object types	Exit criteria
<code>flow.ed_boarding_v1</code>	Encounter, Admit Decision, Placement Request, Bed Stay, Bed, Unit, Transport Job	physical occupancy validated; 95%+ lifecycle coverage; source reconciliation
<code>flow.bed_turn_v1</code>	Bed, Bed Stay, EVS Task, EVS Assignment, Placement Request	dirty/start/complete/ready semantics validated; no discharge proxy
<code>flow.discharge_barrier_s_v1</code>	Encounter, Barrier, Ownership, Service Request, Action, Transport/Medication objects	barrier intervals/ownership validated; overlapping exposure method approved
<code>perio.p.case_pacu_v1</code>	OR Case, OR Suite, PACU Stay/Bay, Staff Assignment, Bed Stay, Transport Job	actual phase/location timestamps and PACU hold definition approved
<code>logistics.transport_v1</code>	Transport Job, Transport Assignment, Encounter, Origin, Destination, Vehicle	pickup and dropoff are actual or explicitly labeled proxies
<code>improvement.action_v1</code>	Signal, Alert, Recommendation, Approval, Action, PDSA, Metric	every alert/action state transition audited and reconciled

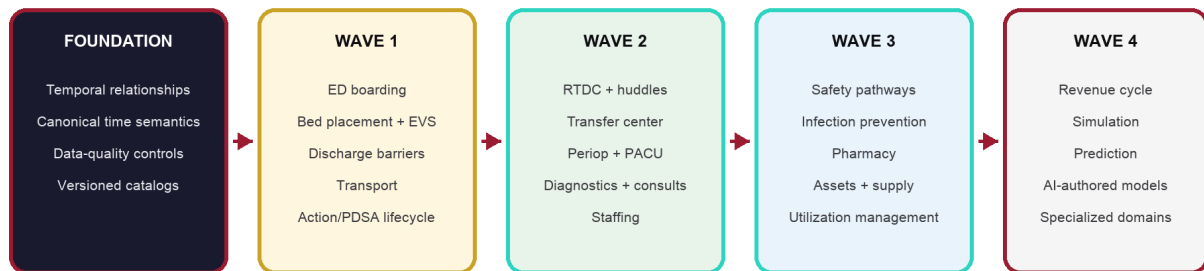
14. Prioritization scorecard

Score each candidate 0–5 on:

- improvement value and strategic pain;
- object-centric necessity (would a single-case log materially distort it?);
- event completeness and timestamp fidelity;
- source-system authority and reconciliation ability;
- actionability and named owner;
- safety/clinical risk of a wrong inference, reverse-scored;
- privacy/workforce sensitivity, reverse-scored;
- cross-domain reuse;
- implementation cost, reverse-scored;
- evidence strength.

14.1 Recommended sequence

Recommended OCEL delivery sequence



Gate every wave on semantic validity, source reconciliation, stakeholder ownership, and measurable actionability.

Figure 3. Recommended delivery waves: semantic and data-quality foundations precede expansion into prediction, simulation, and AI-authored models.

Wave	Models	Why
Foundation	H7 data quality; temporal association objects; canonical time/activity semantics	no downstream model is trustworthy without it
Wave 1	A8 ED boarding; B1/B2 placement/occupancy; B3 EVS turnover; B7 discharge/barriers; E1 transport; H1-H4 improvement lifecycle	strong operational value, clear owners, high object-centric necessity
Wave 2	B10/B11 RTDC/huddles; A9/G8 transfer center/network; C1/C3/C7/C8/C12 periop/PACU; D1/D5/D8 diagnostics/consults; E4 staffing	connects flow to shared services and capacity

Wave	Models	Why
Wave 3	clinical pathway conformance, infection/safety, assets/supply, pharmacy, post-acute, UM	higher governance, data, and exception complexity
Wave 4	revenue cycle, credentialing, procurement, predictive monitoring, simulation, AI-authored models	valuable but less central to first operational proof or more evidence-sensitive

14.2 Do not start with

- a whole-hospital spaghetti map with every object type;
- patient-level longitudinal modeling when encounter-level flow is the actual question;
- clinical conformance based on synthetic or documentation-proxy timestamps;
- worker performance ranking;
- causal “root cause” labels generated from path frequency;
- autonomous intervention;
- simulation before real event semantics and distributions are validated;
- AI narration before deterministic results, provenance, and human review are reliable.

15. Evaluation design for process improvement

Every deployed model should have a charter containing:

- problem statement and operational owner;
- object/event scope and exclusions;
- source contracts and semantic definitions;
- baseline window and seasonality considerations;
- primary outcome measure;
- three to five process measures;
- safety, workforce, equity, and cost balancing measures;
- intervention fidelity measures;
- analysis method and decision thresholds;
- model/data version;
- review cadence and retirement criteria.

Recommended evaluation methods include statistical process control, interrupted time series, stepped-wedge or difference-in-differences where feasible, matched comparisons, and pre-specified prospective protocols. The method must match the intervention and available counterfactual; OCEL provides richer exposure and fidelity data but does not create a counterfactual automatically.

16. Research limitations

- This is a potential-model landscape, not a systematic review with formal database screening and risk-of-bias scoring.
- Direct healthcare OCPM evidence remains sparse; most domain proposals extrapolate from case-centric healthcare process mining and OCEL semantics.
- Many published studies are retrospective, single-center, or method demonstrations.
- Vendor and hospital source semantics vary widely; object and event designs must be validated locally.

- Regulations, standards, measure specifications, and tool support change. Current official specifications and local counsel/governance control.
- OCEL 2.0 represents dynamic attributes but not time-varying O2O validity directly; this report recommends association objects as a modeling convention.
- Process mining can identify patterns and deviations but cannot independently establish clinical cause, safety, or financial ROI.

17. Source register

OCEL and object-centric process mining

- [OCEL 2.0 standard site](#)
- [OCEL 2.0 formal specification](#)
- [Object-Centric Process Mining: Unraveling the Fabric of Real Processes](#)
- [Advancements and Challenges in Object-Centric Process Mining: a systematic literature review](#)
- [OC-PM: analyzing object-centric event logs and process models](#)
- [OPerA: Object-Centric Performance Analysis](#)
- [Object-Centric Local Process Models](#)
- [Toward more realistic simulation models using object-centric process mining](#)
- [Object-centric predictive versus classical predictive monitoring review](#)
- [OCEL 2.0 pm4py support](#)
- [OCEL 2.0 ocpa support](#)
- [Object-centric event-data imperfection patterns](#)

Healthcare OCPM and process mining

- [Park, Lee, and Cho: OMOP CDM to OCEL, MIMIC-IV](#)
- [Celik: Object-Centric Process Mining of Laparoscopic Cholecystectomy Workflows](#)
- [Aversano et al.: healthcare process-mining systematic review](#)
- [Santos et al.: healthcare process-mining tertiary study](#)
- [Process mining for healthcare: characteristics and challenges](#)
- [ED performance analysis](#)
- [Pediatric trauma patient-flow mapping](#)
- [Perioperative EHR workflow study](#)
- [Operating-room RTLS/process-mining dashboard study](#)
- [Hospital patient-transport process-mining study](#)
- [Clinical chemistry laboratory process-mining study](#)
- [Red-blood-cell wastage process-mining study](#)
- [Hospital pathway/resource study across inpatient, outpatient, lab, and radiology](#)
- [Stroke time-perspective process mining](#)
- [Privacy-Preserving Process Mining in Healthcare](#)

Hospital improvement, interoperability, and governance

- [IHI: Achieving Hospital-wide Patient Flow, second edition](#)
- [CMS hospital QAPI condition](#)

- [HL7 FHIR R5 workflow](#)
- [HL7 FHIR R5 ServiceRequest](#)
- [OMOP CDM](#)
- [HHS HIPAA de-identification guidance](#)

Final recommendation

Adopt OCEL 2.0 as Zephyrus's canonical process-intelligence exchange contract, but treat the **semantic model and source contracts** as more important than the file format. Build the first production proof around ED boarding, placement/physical occupancy, EVS bed turnover, discharge barriers, transport, and the action/PDSA lifecycle. These models have clear operational ownership, strong object interaction, measurable outcomes, and direct alignment with Zephyrus's hospital-operations thesis.

Do not expand the object catalog faster than source semantics can be validated. A smaller, temporally correct log that distinguishes real physical events from documentation proxies will produce more improvement value than a hospital-wide catalog filled with ambiguous timestamps and timeless relationships.